

ECEN 215, Spring 2024

Electrical Circuit Theory

Class 1: Intro. To EE

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OUTLINE

- Introduction to Electrical Engineering
- Introduction to DC Circuits



Overview of Electrical Engineering



Two main functions of engineering

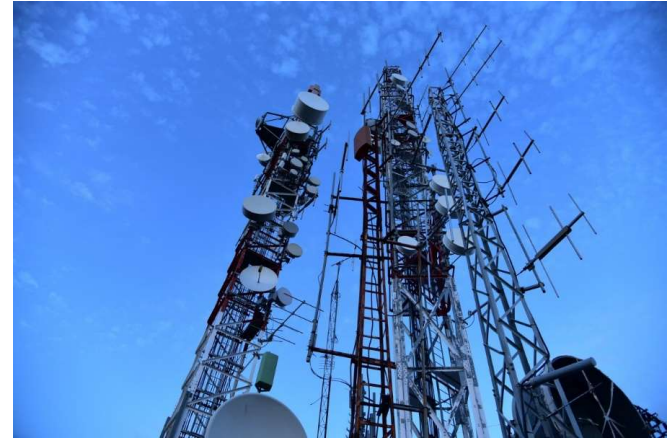
- **Analysis** – break it down
- **Design** – build it up

What's electricity useful for?

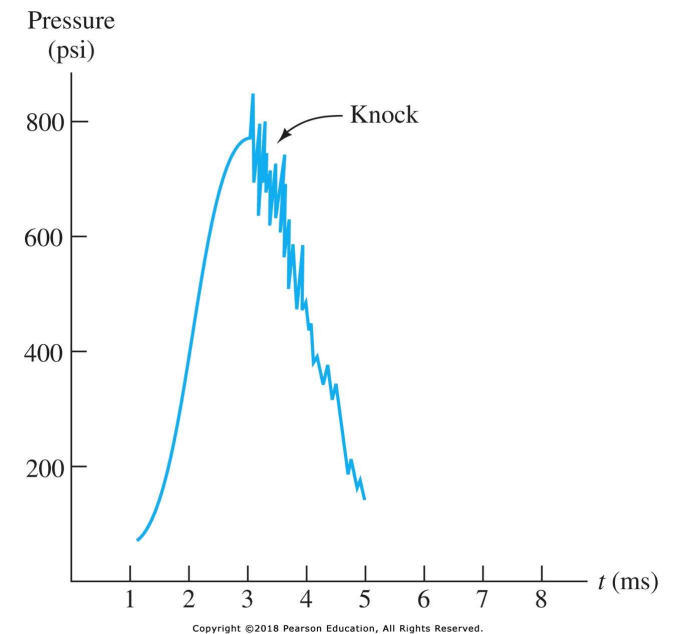
- Transmitting **energy** – the ability to do useful work
- Transmitting **information** – data with meaning

What makes electricity especially good for that?

- It is **fast** – we can send vast amounts of energy and information at nearly the speed of light across continents (and information across oceans and space)
- It is **efficient** – very little energy losses in conductors, information can be processed in nano-sized devices



Overview of Electrical Engineering



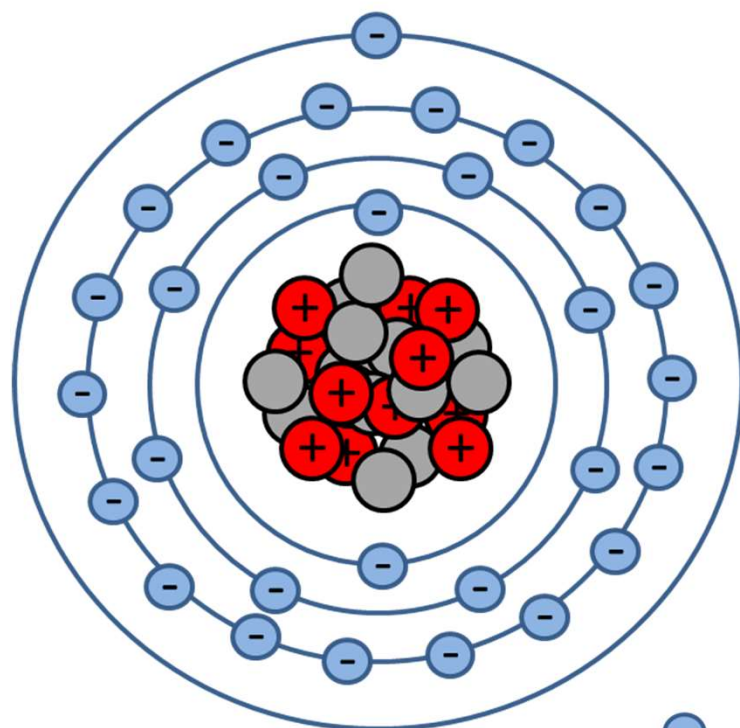
Pressure versus time for an internal combustion engine experiencing knock. Sensors convert pressure to an electrical signal, Electrical circuit can process this data



What is a Charge?

- Electric charge is the basis of describing all electrical phenomenon.
- Charge is bipolar (it can be positive or negative)
- Charge is discrete
 - Each electron carries a charge of $1.6022 \times 10^{-19}\text{C}$.
 - 1 Coulomb = 6.24×10^{19} electrons.
- Separation of charge creates an electric force
 - Voltage (Volts)
- Motion of charge creates an electric current
 - Current (Amps)

What is Electricity?

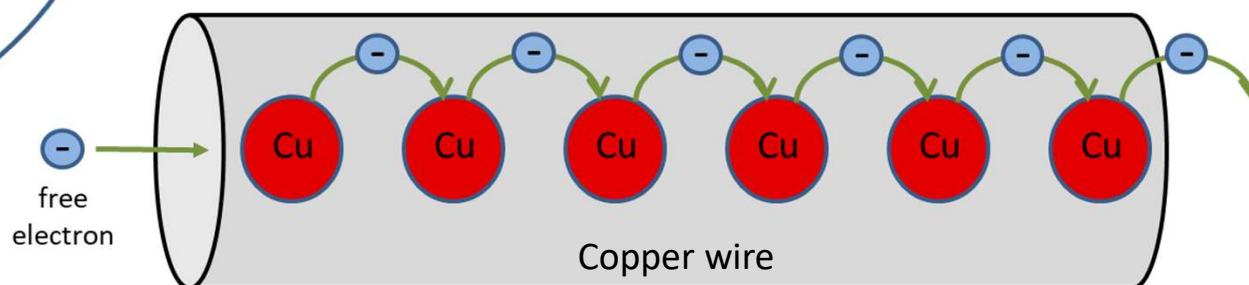


Copper atom (Cu)

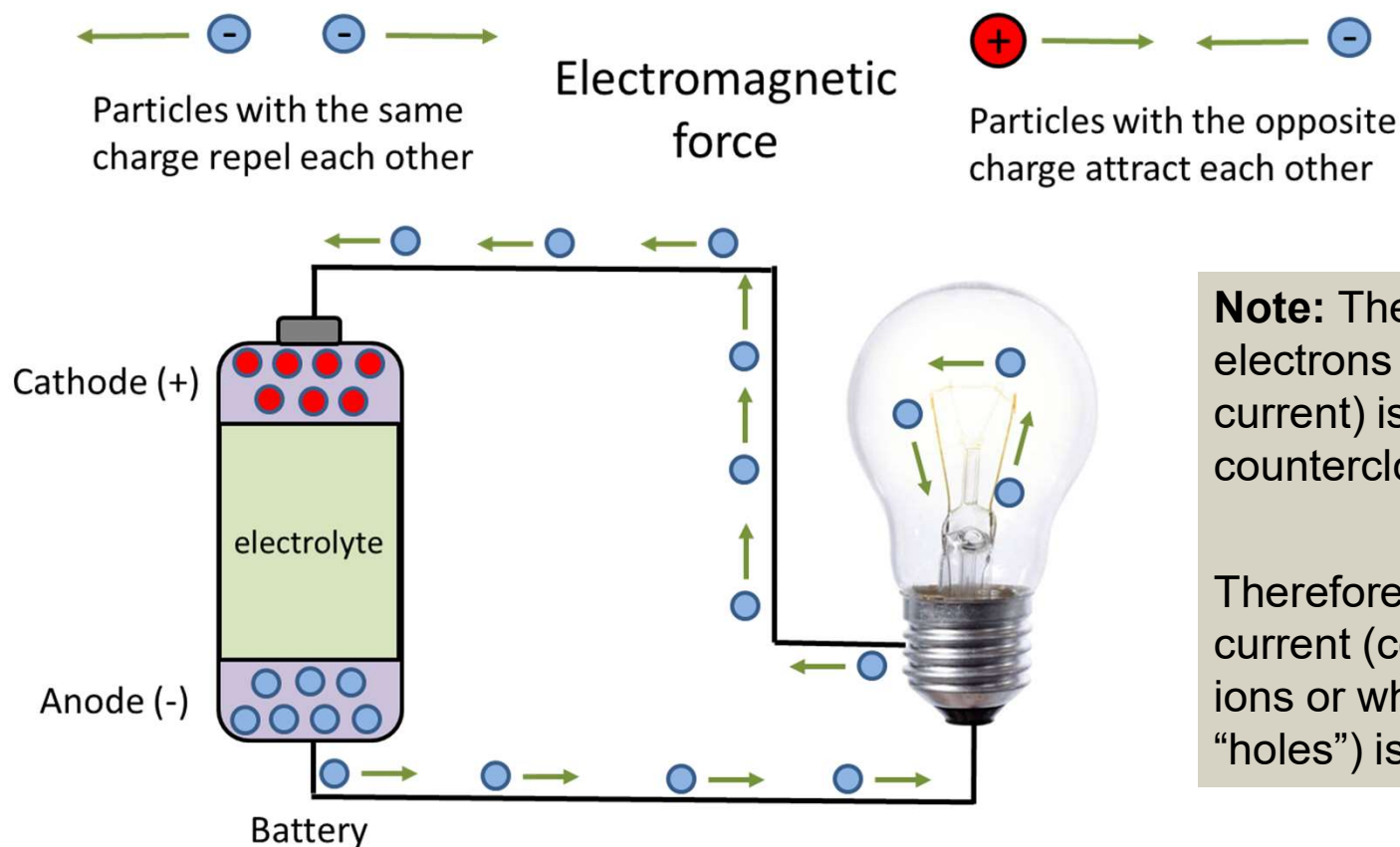
- 29 protons (+ charge)
- 35 Neutrons (neutral charge)
- 29 electrons (- charge)

Single electron in the outer shell makes it relatively easy to dislodge an electron from the CU atom.

⇒ good conductor of electricity



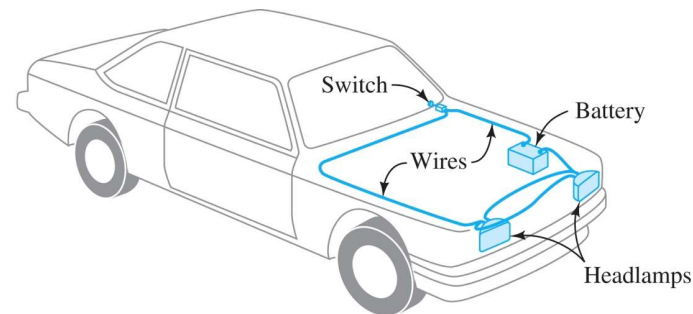
How Do We Make Electrons Move?



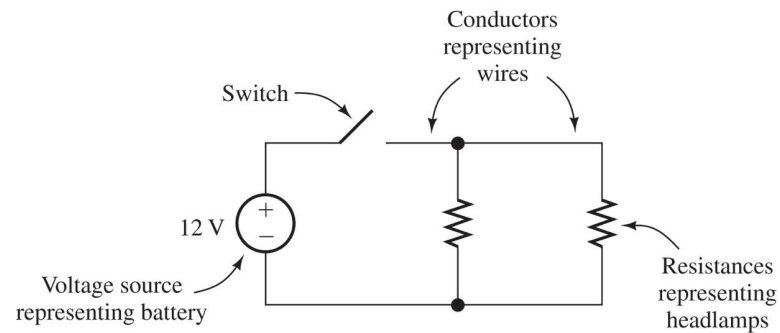
Note: The physical flow of electrons (negatively charged current) is moving counterclockwise in this circuit.

Therefore, positively charged current (consisting of positive ions or what we sometimes call "holes") is flowing clockwise.

Electric Circuit



(a) Physical configuration



(b) Circuit diagram

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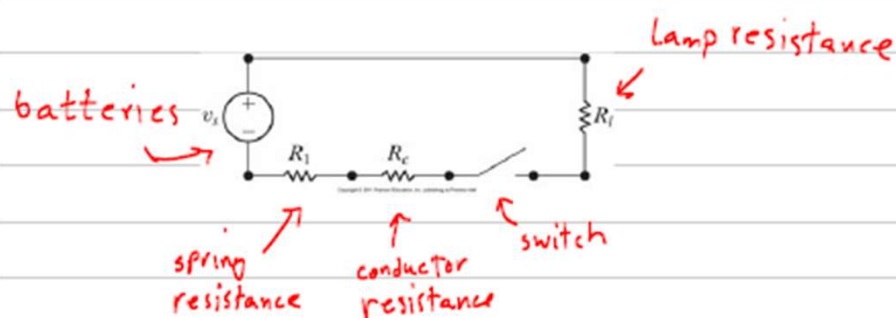
The headlight circuit. (a) The actual physical layout of the circuit. (b) The circuit diagram.

Electric Circuit Models



The circuit elements are used to construct the circuit model for practical systems.

Example: circuit model of a flashlight



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Next Lecture

- **Current, Voltage, and Power**

